

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT2-Constraint-Based Problem Solving

Register Number	
Name	

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 20%

QUESTIONS:3

Total Marks:30

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- Analyze all the given constraints and explain how they affect the robot's decision-making process.
- Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

Question 1: Hospital Scheduling using Backtracking Search

1.1 Problem Understanding

4. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- vi. D1 cannot work Night shift
- vii. D2 must work before D3 (Morning < Afternoon < Night)
- viii. Only one doctor per shift
- ix. D3 cannot work Morning
- x. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

1.2 Constraint Analysis

- $D1 \neq \text{Night}$
- D2 before D3
- One doctor per shift
- $D3 \neq \text{Morning}$
- Every doctor gets exactly one shift

1.3 Search Space

Doctors = {D1,D2,D3}

Shifts = {Morning,Afternoon,Night}

1.4 Backtracking Search Steps

Step 1

Assign

D1 = Morning

Valid

Step 2

Assign

D2 = Afternoon

Valid

Step 3

Assign

D3 = Night

Check Constraints

D1 ≠ Night
D2 before D3
One doctor per shift
D3 ≠ Morning
All assigned

Valid Solution Found.

Final Schedule

Doctor	Shift
--------	-------

D1	Morning
----	---------

D2	Afternoon
----	-----------

D3	Night
----	-------

Question 2: Robot Navigation using BFS

2.1 Problem Statement:

A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- v. Movement allowed: Up, Down, Left, Right
- vi. Cost of each move = 1
- vii. Cannot pass through obstacles
- viii. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

Goal: To Find shortest path from Start to Goal.

2.2 Constraints

- Move Up
- Move Down

- Move Left
- Move Right
- Obstacle Avoidance
- Cost = 1

2.3 Grid Representation

Use:

```
S 0 0 X 0
0 X 0 X 0
0 X 0 0 0
0 0 X X 0
0 0 0 G 0
```

Where

S = Start

G = Goal

X = Obstacle

2.4 BFS Node Expansion

Expand S

Queue:

[(1,0),(0,1)]

Expand (1,0)

Queue:

[(0,1),(2,0)]

Expand (0,1)

Queue:

[(2,0),(0,2)]

Continue until Goal.

2.5 Optimal Path

(0,0),(1,0),(2,0),(3,0),(4,0),(4,1),(4,2),(4,3)

Goal Reached.

Total Cost

7 Moves

Cost = 7×1

Cost = 7

Question 3: Autonomous Rescue Robot

a) Constraint Analysis

The robot can move only in four directions. This limits the available actions at each state.

Some cells contain obstacles. These cells cannot be traversed and must be avoided.

The robot can observe only adjacent cells. Therefore, the environment is partially observable.

Every movement has a cost of 1. The robot must minimize unnecessary movements.

Risky zones introduce an additional cost of 2. The robot should avoid risky areas whenever possible.

The robot continuously updates its knowledge while exploring. Therefore, decisions must be revised dynamically.

b) Solution Approach

An Online Search Agent combined with Uniform Cost Search is an appropriate solution.

Initially, the robot knows only its current location.

The robot observes adjacent cells and identifies safe and blocked paths.

The robot updates its internal map after every movement.

The robot calculates the cumulative path cost.

Safe paths with lower costs are preferred over risky paths.

The process continues until the survivor location G is reached.

c) Application of AI Concepts

Intelligent Agent

The robot acts as an intelligent agent because it perceives the environment through sensors and performs actions based on the information received.

Rationality

The robot is rational because it selects actions that minimize total cost while maximizing safety and the probability of reaching the survivor.

Environment Type

The environment is partially observable because the robot cannot view the entire building.

The environment is dynamic because new information becomes available during movement.

The environment is sequential because each decision influences future decisions.

The environment is uncertain because obstacles and risky zones may not be known initially.

d) Justification of the Chosen Approach

The Online Search Agent is the most suitable approach because the robot does not have complete information about the environment.

Uniform Cost Search is effective because movement costs and risk costs vary.

The method guarantees that the robot chooses the lowest-cost safe path.

The approach supports continuous learning and dynamic decision-making, making it ideal for disaster rescue operations in partially observable environments.